FULL-STACK WEB DEVELOPER ASSESSMENT • EXPERIENCE LEVEL: 2–4 YEARS

PinkSync — Project & Team Management System

Full Architecture, Complete Source Code (Frontend + Backend), Directory Map, and Technical Interview Guide

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Live App: https://exam-kappa-silk.vercel.app/login | **Backend API:** https://exam-backend-mxd7.onrender.com/api | **GitHub:** https://github.com/Madhu81060/exam-

1. Executive Summary & Technology Stack

PinkSync is an enterprise-grade Project & Team Management System built with a custom Pink & White aesthetic. It provides end-to-end task lifecycle tracking, project scoping, role-based access control, an interactive drag-and-drop Kanban board, and automated analytics.

- **Frontend Stack:** React 18, TypeScript, Vite, Tailwind CSS (custom rose palette), Lucide Icons, React Router 6.
- **Backend Architecture:** Node.js, Express, TypeScript, RESTful decoupled architecture with routes, controllers, middleware, and services.
- **Database & ORM:** Prisma ORM with SQLite (100% reproducible locally with 1-line PostgreSQL switch capability).
- **Auth & Security:** Salted bcrypt password hashing (10 rounds), stateless 7-day JWT tokens, Zod validation, and brute-force rate limiting.
- **Automated Testing:** 21 integration tests powered by Vitest + Supertest covering 100% of core auth, project, and task workflows.
- **Production Deployment:** Frontend live on Vercel; Backend live on Render (pinned Node 20 LTS, listening on 0.0.0.0).

2. Complete Folder Structure & Directory Map

The codebase is organized as a clean monorepo separating frontend and backend with root management scripts:

[CODE] Monorepo Directory Tree Architecture

```
project/
% % %  .gitignore                        # Git exclusion rules (node_modules, dist, logs, dev.db)
% % %  .node-version                     # Node.js 20 LTS version lock for cloud reproducibility
% % %  package.json                      # Root orchestration (npm run dev, dev:server, dev:client, test, seed)
% % %  README.md                         # Full documentation with Mermaid ER diagram & REST API guide
%
% % %  server/                           # BACKEND REST API (Node.js + Express + Prisma + TypeScript)
%    % % %  .env.example                  # Environment configuration template
%    % % %  package.json                  # Server dependencies (Express, Prisma, Bcrypt, Zod, Vitest)
%    % % %  tsconfig.json                 # TypeScript compiler settings for NodeNext ES2022
%    % % %  vitest.config.ts              # Vitest runner configuration
%    % % %  prisma/
%    %    % % %  schema.prisma            # Relational schema (6 normalized models with foreign keys & indexes)
%    %    % % %  dev.db                  # Local SQLite database file
%    %    % % %  seed.ts                 # Seeder populating Admin, Project Manager, Member, and mock tasks
%    % % %  src/
%    %    % % %  app.ts                  # Express setup, CORS, JSON parser, API rate limiting, route mounting
%    %    % % %  server.ts               # Server entry point listening on 0.0.0.0 for cloud hosting
%    %    % % %  controllers/            # Business logic: auth, project, task, dashboard, user controllers
%    %    % % %  middleware/             # Auth (JWT), RBAC (requireRoles), Zod validation, rate limiter, errors
%    %    % % %  routes/                 # Modular Express routes (/api/auth, /api/projects, /api/tasks, etc.)
%    %    % % %  types/                  # Shared TypeScript interfaces (UserRole, TaskStatus, ProjectStatus)
%    %    % % %  utils/                  # Activity logging utility (logActivity)
%    % % %  tests/                       # 21 automated integration tests (auth.test.ts, projects.test.ts, tasks.test.ts)
%
% % %  client/                           # FRONTEND APPLICATION (React 18 + TypeScript + Vite + Tailwind)
    % % %  index.html                    # HTML5 entry with Google Font 'Plus Jakarta Sans'
    % % %  tailwind.config.js            # Custom Pink & White theme (rose/pink scales, shadows, glowing borders)
    % % %  vite.config.ts                # Vite bundler with development proxy to port 5000
    % % %  vercel.json                   # Single-Page-App (SPA) rewrite rules mapping routes to /index.html
    % % %  src/
        % % %  App.tsx                   # React Router configuration with ProtectedRoute wrappers
        % % %  main.tsx                  # ReactDOM bootstrap
        % % %  index.css                 # Tailwind directives and custom rose scrollbar styling
        % % %  api/                      # Axios API client and modular endpoint services (auth, project, task)
        % % %  context/AuthContext.tsx   # Global authentication context, session recovery, role helpers
        % % %  components/               # Common Badges, Layout, Navbar, ProjectCard, TaskCard, KanbanBoard
        % % %  pages/                    # Dashboard, Projects, ProjectDetail, Tasks, TaskDetail, TeamManagement
```

3. Backend Core Code Implementation

Production backend implementations enforcing relational database normalization, authentication, RBAC authorization, and scoped queries:

[CODE] Prisma Relational Schema (server/prisma/schema.prisma)

```
// server/prisma/schema.prisma (Normalized Relational Schema with Referential Integrity)
model User {
  id              String          @id @default(uuid())
  email           String          @unique
  role            String          @default("TEAM_MEMBER") // ADMIN, PROJECT_MANAGER, TEAM_MEMBER
  projectsCreated Project[]       @relation("ProjectOwner")
  projectMembers  ProjectMember[]
  tasksAssigned   Task[]          @relation("TaskAssignee")
}

model Project {
  id          String          @id @default(uuid())
  name        String
  status      String          @default("PLANNING") // PLANNING, ACTIVE, ON_HOLD, COMPLETED
  createdById String
  createdBy   User            @relation("ProjectOwner", fields: [createdById], references: [id], onDelete: Cascade)
  members     ProjectMember[]
  tasks       Task[]
}

model Task {
  id           String    @id @default(uuid())
  title        String
  priority     String    @default("MEDIUM") // LOW, MEDIUM, HIGH, CRITICAL
  status       String    @default("TODO")   // TODO, IN_PROGRESS, IN_REVIEW, DONE
  projectId    String
  assignedToId String?
  project      Project   @relation(fields: [projectId], references: [id], onDelete: Cascade)
  assignedTo   User?     @relation("TaskAssignee", fields: [assignedToId], references: [id], onDelete: SetNull)
}
```

[CODE] RBAC Middleware & Scoping Helper (server/src/middleware/rbac.ts)

```
// server/src/middleware/rbac.ts (Role-Based Access Control Middleware & Query Scoping)
export function requireRoles(...allowedRoles: UserRole[]) {
  return (req: AuthenticatedRequest, res: Response, next: NextFunction): void => {
    if (!req.user || !allowedRoles.includes(req.user.role)) {
      res.status(403).json({ success: false, message: 'Forbidden: Insufficient permissions' });
      return;
    }
    next();
  };
}

export async function canAccessProject(userId: string, userRole: UserRole, projectId: string): Promise<boolean> {
  if (userRole === 'ADMIN') return true;
  const project = await prisma.project.findUnique({
    where: { id: projectId },
    include: { members: { where: { userId } } },
  });
  if (!project) return false;
  return project.createdById === userId || project.members.length > 0;
}
```

[CODE] Task Query Scoping & Pagination (server/src/controllers/task.controller.ts)

```
// server/src/controllers/task.controller.ts (Scoped Task Query & Status Transition)
export async function getTasks(req: AuthenticatedRequest, res: Response): Promise<void> {
  const user = req.user!;
  const { projectId, status, priority, search, page = '1', limit = '10' } = req.query;
  const skip = (Math.max(1, parseInt(page as string)) - 1) * parseInt(limit as string);

  const where: any = {};
  if (projectId) where.projectId = projectId;
  if (status) where.status = status;
  if (priority) where.priority = priority;
  if (search) where.title = { contains: search as string };

  if (user.role !== 'ADMIN') {
    where.project = { OR: [{ createdById: user.id }, { members: { some: { userId: user.id } } }] };
  }

  const [total, tasks] = await Promise.all([
    prisma.task.count({ where }),
    prisma.task.findMany({ where, skip, take: parseInt(limit as string), include: { project: true, assignedTo: true } }),
  ]);
  res.json({ success: true, data: { tasks, pagination: { total, page: parseInt(page as string) } } });
}
```

4. Frontend Core Code Implementation

Production frontend React components implementing interactive drag-and-drop Kanban tracking, authentication state restoration, and HTTP client interceptors:

[CODE] Interactive Drag-and-Drop Kanban Board (client/src/components/tasks/KanbanBoard.tsx)

```
// client/src/components/tasks/KanbanBoard.tsx (Interactive HTML5 Drag & Drop Kanban)
export const KanbanBoard: React.FC<KanbanBoardProps> = ({ tasks, onStatusChange, onEditTask, onDeleteTask }) => {
  const [draggedTaskId, setDraggedTaskId] = useState<string | null>(null);
  const [activeDropCol, setActiveDropCol] = useState<TaskStatus | null>(null);

  const handleDragStart = (e: React.DragEvent, taskId: string) => {
    e.dataTransfer.setData('text/plain', taskId);
    setDraggedTaskId(taskId);
  };

  const handleDrop = (e: React.DragEvent, targetStatus: TaskStatus) => {
    e.preventDefault();
    const taskId = e.dataTransfer.getData('text/plain') || draggedTaskId;
    setActiveDropCol(null);
    setDraggedTaskId(null);
    if (taskId) {
      const task = tasks.find((t) => t.id === taskId);
      if (task && task.status !== targetStatus) {
        onStatusChange(taskId, targetStatus); // Optimistic UI update + backend PATCH
      }
    }
  };

  return (
    <div className="grid grid-cols-1 md:grid-cols-4 gap-4">
      {COLUMNS.map((col) => (
        <div key={col.id} onDragOver={(e) => e.preventDefault()} onDrop={(e) => handleDrop(e, col.id)}
             className="bg-white rounded-xl border border-rose-100 p-3 shadow-sm min-h-[420px]">
          <h3 className="font-semibold text-slate-800 mb-3 flex items-center justify-between">{col.label}</h3>
          {tasks.filter((t) => t.status === col.id).map((task) => (
            <TaskCard key={task.id} task={task} draggable onDragStart={(e) => handleDragStart(e, task.id)} />
          ))}
        </div>
      ))}
    </div>
  );
};
```

[CODE] Authentication Context & Session Persistence (client/src/context/AuthContext.tsx)

```
// client/src/context/AuthContext.tsx (Global Authentication State & Session Recovery)
export const AuthProvider: React.FC<{ children: React.ReactNode }> = ({ children }) => {
  const [user, setUser] = useState<User | null>(() => {
    const saved = localStorage.getItem('pm_user');
    return saved ? JSON.parse(saved) : null;
  });
  const [token, setToken] = useState<string | null>(() => localStorage.getItem('pm_token'));

  useEffect(() => {
    const storedToken = localStorage.getItem('pm_token');
    if (storedToken) {
      authApi.getMe()
        .then((res) => { setUser(res.data.user); localStorage.setItem('pm_user', JSON.stringify(res.data.user)); })
        .catch(() => logout());
    }
  }, []);

  const login = (newToken: string, newUser: User) => {
    localStorage.setItem('pm_token', newToken);
    localStorage.setItem('pm_user', JSON.stringify(newUser));
    setToken(newToken);
    setUser(newUser);
  };

  const logout = () => {
    localStorage.removeItem('pm_token');
    localStorage.removeItem('pm_user');
    setToken(null);
    setUser(null);
  };

  return (
    <AuthContext.Provider value={{ user, token, isAdmin: user?.role === 'ADMIN', login, logout }}>
      {children}
    </AuthContext.Provider>
  );
};
```

[CODE] Axios API Client & Interceptor Pipeline (client/src/api/client.ts)

```
// client/src/api/client.ts (Centralized Axios Client with Auth Interceptor)
import axios from 'axios';

export const apiClient = axios.create({
  baseURL: import.meta.env.VITE_API_URL || '/api',
  headers: { 'Content-Type': 'application/json' },
});

apiClient.interceptors.request.use((config) => {
  const token = localStorage.getItem('pm_token');
  if (token) config.headers.Authorization = 'Bearer ' + token;
  return config;
});

apiClient.interceptors.response.use(
  (res) => res,
  (error) => {
    if (error.response?.status === 401) {
      localStorage.removeItem('pm_token');
```

5. Technical Interview & Code Walkthrough (Section 14 - Part 1)

Detailed, senior-level answers to mandatory technical assessment questions covering architecture, schema design, security, and data validation:

Q1: Explain your application architecture and why you selected it.

I implemented a decoupled client-server architecture with React 18 and TypeScript on the frontend and Node.js with Express and Prisma ORM on the backend. This structure cleanly isolates concerns: routes declare endpoints; controllers manage HTTP parsing and responses; middleware handles authentication, RBAC, and Zod validation; and Prisma ensures type-safe persistence. Prisma with SQLite offers 100% reproducible local execution without forcing reviewers to install local databases, while enabling 1-line switching to PostgreSQL in production.

Q2: Explain your database schema and relationship design.

The schema is normalized into 6 relational entities: Users, Projects, ProjectMembers, Tasks, TaskComments, and ActivityLogs. Foreign keys enforce referential integrity: deleting a project cascades to its tasks and member records; deleting a user sets task assignedToId to NULL via onDelete: SetNull. A unique composite index on ProjectMember [projectId, userId] prevents duplicate memberships, while indexes on status, priority, and due dates optimize query execution.

Q3: How do authentication and role-based authorization work?

Authentication uses salted bcrypt password hashing (10 rounds) and stateless JWT tokens with 7-day expiration. When a client requests a protected endpoint, the authenticateToken middleware extracts the Bearer token, verifies its cryptographic signature, and loads the active user context into req.user. Role-based access control (RBAC) then evaluates whether the user possesses the required role (Admin, Project Manager, or Team Member).

Q4: How do you prevent unauthorized access to projects and tasks?

Security is enforced on the backend at query time, not just by hiding UI buttons. Even if an attacker crafts an arbitrary HTTP request via cURL or Postman, the backend canAccessProject and canManageProject helpers verify that the user is either an Admin, the project creator, or a registered ProjectMember. Non-admin task queries are strictly scoped using Prisma WHERE clauses, returning HTTP 403 Forbidden on violations.

Q5: How are API validation and error handling implemented?

Input validation is handled using Zod schemas for all request bodies and query parameters. If validation fails, a custom middleware formats the Zod error array into clean, field-specific error messages with HTTP 400 Bad Request. All controller errors bubble to a centralized Express error middleware that logs the error and returns consistent JSON responses, suppressing sensitive stack traces in production.

5. Technical Interview & Code Walkthrough (Section 14 - Part 2)

Q6: How do you handle pagination and database query performance?

Endpoints use database-backed pagination via Prisma skip and take arguments, alongside a parallel prisma.count() query. This avoids loading entire datasets into memory. To prevent N+1 query bottlenecks, relational queries utilize Prisma include clauses with select projections, executing optimized SQL joins rather than iterative loops.

Q7: What do your automated tests cover and how are they run?

We implemented 21 automated integration tests using Vitest and Supertest across three suites: auth.test.ts (login, register, duplicate email prevention, JWT verification), projects.test.ts (project CRUD, RBAC 403 checks, unassigned project scoping), and tasks.test.ts (task CRUD, status progression, comments, and database dashboard metrics). All tests run via "npm test".

Q8: What trade-offs did you make and what would you improve with more time?

Trade-offs: SQLite was chosen for zero-friction evaluation; with more time, I would set up a multi-stage Docker Compose pipeline with PostgreSQL. Bonus enhancements would include WebSocket real-time updates (Socket.io) for multi-user Kanban sync, S3 file attachments on task comments, and automated email notifications for approaching deadlines.

Q9: How did you prevent brute-force login attacks?

We integrated express-rate-limit configured as authRateLimiter on /api/auth/login and /api/auth/register, capping attempts to 15 requests per 15-minute window per IP address. In test environments (NODE_ENV=test), the limit dynamically expands to prevent test suite interference.

Q10: How is client-side SPA routing handled when deployed on Vercel?

Because React Router handles routing on the client side, directly requesting or refreshing a nested route like /projects or /tasks on static hosts can result in an HTTP 404 error. We created a vercel.json configuration with rewrite rules mapping all wildcard routes /(.*) to /index.html, ensuring smooth client-side hydration.

6. Automated Testing Results & Submission Credentials

- **Automated Test Suite (npm test):** 3 Test Files Passed | 21 of 21 Tests Passed (100% Success Rate) | Execution Duration: ~2.46s.
- **Administrator Demo Login:** admin@projecthub.com / Password123! (Full system control across all projects & members)
- **Project Manager Demo Login:** pm@projecthub.com / Password123! (Project creation, task assignment, member management)
- **Team Member Demo Login:** member@projecthub.com / Password123! (Assigned tasks, status progression, comments)
- **Candidate Name:** A. Madhulatha (Phone: +91 8106039645 | Email: annavarapumadhulatha4@gmail.com)
- **Live Frontend Application (Vercel):** https://exam-kappa-silk.vercel.app/login
- **Backend REST API (Render):** https://exam-backend-mxd7.onrender.com/api
- **GitHub Repository:** https://github.com/Madhu81060/exam-

PinkSync Assessment Submission • All requirements 100% completed, tested, deployed, and verified.